Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Factorising Quadratics

Name _____ Date _____ No calculator Version 1

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KEY WORDS factorise *(whakatauwehe)* · factor *(tauwehe)* · common factor *(tauwehe pātahi)* · quadratic *(pūrua)*

A Common factors ●○○

Remember: Factorising is the reverse of expanding. Take out the largest factor that goes into every term.

1. Factorise fully.

a) $6x + 9$ = _____ **b)** $5x - 20$ = _____ **c)** $12x + 8$ = _____ **d)** $14x - 7$ = _____

e) $8x - 16$ = _____ **f)** $12x + 30$ = _____ **g)** $x^2 + 5x$ = _____ **h)** $4x^2 - 6x$ = _____

i) $3x^2 + 12x$ = _____ **j)** $15x^2 - 10x$ = _____ **k)** $7x^2 - 21x$ = _____ **l)** $12x^2 - 20x$ = _____

B Factorising x² + bx + c ●●○

Remember: Look for two numbers that multiply to give the last number and add to give the middle number.

2. Factorise.

a) $x^2 + 5x + 6$ **b)** $x^2 + 7x + 6$ **c)** $x^2 + 9x + 20$

d) $x^2 + 10x + 16$ **e)** $x^2 + 13x + 30$ **f)** $x^2 - 5x + 6$

g) $x^2 - 8x + 7$ **h)** $x^2 - 10x + 24$ **i)** $x^2 - 13x + 40$

j) $x^2 + 3x - 10$ **k)** $x^2 + 4x - 21$ **l)** $x^2 - 6x - 16$

m) $x^2 - 5x - 36$ **n)** $x^2 + 9x - 10$ **o)** $x^2 - 9x - 36$

C Difference of two squares ●●○

Remember: $a^2 - b^2 = (a + b)(a - b)$

3. Factorise.

a) $x^2 - 9$ **b)** $x^2 - 25$ **c)** $x^2 - 1$

d) $x^2 - 100$ **e)** $x^2 - 49$ **f)** $4x^2 - 9$

g) $9x^2 - 16$ **h)** $16x^2 - 1$ **i)** $25x^2 - 4$

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D Perfect squares ●●○

4. Factorise.

a) $x^2 + 6x + 9$ **b)** $x^2 - 10x + 25$ **c)** $x^2 + 14x + 49$

d) $x^2 - 2x + 1$ **e)** $x^2 + 20x + 100$ **f)** $x^2 - 18x + 81$

g) $4x^2 + 4x + 1$ **h)** $9x^2 - 12x + 4$ **i)** $25x^2 + 10x + 1$

E Mixed practice ●●●

⚠ **Watch out:** Always look for a common factor first. Then check whether what is left can be factorised again.

5. Factorise fully. Decide which method to use first.

a) $x^2 + 10x + 21$ **b)** $x^2 - 64$ **c)** $8x^2 - 20x$

d) $x^2 - 8x + 16$ **e)** $x^2 - x - 30$ **f)** $2x^2 + 10x + 12$

g) $16x^2 - 1$ **h)** $3x^2 - 9x - 12$ **i)** $x^2 - 13x + 22$

j) $5x^2 - 20$ **k)** $x^2 + 18x + 81$ **l)** $4x^2 + 8x + 4$

m) $6x + 24$ **n)** $x^2 + 7x - 60$ **o)** $2x^2 - 50$

F Problems ●●●

6. A square has an area of $x^2 + 10x + 25$.

a) Find an expression for the length of its sides. **b)** Find an expression for its perimeter.

7. A rectangle has an area of $x^2 + 7x + 12$ and a width of $x + 3$. Find an expression for its length.

8. Kiri says that $x^2 - 5x + 6 = (x - 2)(x + 3)$. Expand her answer to show that she is wrong, then factorise correctly.

★ Extension

9. Use the difference of two squares to work out $51^2 - 49^2$ without a calculator.

10. Factorise $(x + 1)^2 - 4$.

11. Factorise $x^4 - 16$ fully.

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1-1 PRIVATE TUTORS

ALGEBRA

Factorising Quadratics

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS factorise *(whakatauwehe)* · factor *(tauwehe)* · common factor *(tauwehe pātahi)* · quadratic *(pūrua)*

A Common factors ●○○

Remember: Factorising is the reverse of expanding. Take out the largest factor that goes into every term.

1. Factorise fully.

a) $6x + 10$ = _____ **b)** $6x - 18$ = _____ **c)** $10x + 15$ = _____ **d)** $18x - 9$ = _____

e) $8x - 24$ = _____ **f)** $8x + 28$ = _____ **g)** $x^2 + 7x$ = _____ **h)** $6x^2 - 15x$ = _____

i) $4x^2 + 12x$ = _____ **j)** $10x^2 - 6x$ = _____ **k)** $6x^2 - 30x$ = _____ **l)** $15x^2 - 20x$ = _____

B Factorising x² + bx + c ●●○

Remember: Look for two numbers that multiply to give the last number and add to give the middle number.

2. Factorise.

a) $x^2 + 5x + 4$ **b)** $x^2 + 7x + 10$ **c)** $x^2 + 9x + 18$

d) $x^2 + 11x + 18$ **e)** $x^2 + 14x + 40$ **f)** $x^2 - 6x + 5$

g) $x^2 - 6x + 8$ **h)** $x^2 - 11x + 24$ **i)** $x^2 - 13x + 42$

j) $x^2 + 4x - 12$ **k)** $x^2 + 5x - 24$ **l)** $x^2 - 4x - 21$

m) $x^2 - 8x - 20$ **n)** $x^2 + 8x - 9$ **o)** $x^2 - 8x - 48$

C Difference of two squares ●●○

Remember: $a^2 - b^2 = (a + b)(a - b)$

3. Factorise.

a) $x^2 - 16$ **b)** $x^2 - 36$ **c)** $x^2 - 4$

d) $x^2 - 81$ **e)** $x^2 - 64$ **f)** $4x^2 - 25$

g) $9x^2 - 4$ **h)** $16x^2 - 9$ **i)** $25x^2 - 1$

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D Perfect squares ●●○

4. Factorise.

a) $x^2 + 8x + 16$ **b)** $x^2 - 12x + 36$ **c)** $x^2 + 16x + 64$

d) $x^2 - 4x + 4$ **e)** $x^2 + 22x + 121$ **f)** $x^2 - 14x + 49$

g) $4x^2 + 12x + 9$ **h)** $9x^2 - 6x + 1$ **i)** $16x^2 + 8x + 1$

E Mixed practice ●●●

⚠ **Watch out:** Always look for a common factor first. Then check whether what is left can be factorised again.

5. Factorise fully. Decide which method to use first.

a) $x^2 + 10x + 24$ **b)** $x^2 - 49$ **c)** $9x^2 - 12x$

d) $x^2 - 6x + 9$ **e)** $x^2 - 3x - 28$ **f)** $2x^2 + 10x + 8$

g) $25x^2 - 1$ **h)** $3x^2 - 9x - 30$ **i)** $x^2 - 13x + 30$

j) $4x^2 - 36$ **k)** $x^2 + 20x + 100$ **l)** $5x^2 + 10x + 5$

m) $7x + 21$ **n)** $x^2 + 7x - 44$ **o)** $3x^2 - 48$

F Problems ●●●

6. A square has an area of $x^2 + 12x + 36$.

a) Find an expression for the length of its sides. **b)** Find an expression for its perimeter.

7. A rectangle has an area of $x^2 + 7x + 10$ and a width of $x + 2$. Find an expression for its length.

8. Kiri says that $x^2 - 5x + 4 = (x - 1)(x + 4)$. Expand her answer to show that she is wrong, then factorise correctly.

★ Extension

9. Use the difference of two squares to work out $76^2 - 24^2$ without a calculator.

10. Factorise $(x + 2)^2 - 9$.

11. Factorise $x^4 - 81$ fully.

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